

Bachelor of Technology — Mechanical Engineering

Table 1: Mechanical Engineering — B23 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	50
	Discipline Electives	16
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 2: B. Tech Mechanical Engineering — B23 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EE-261	Electrical Systems Around Us	3-0-0-3	3
2	IC-241	Material Science for Engineers	3-0-0-3	3
3	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
4	ME-210	Fluid Mechanics	2.5-0.5-0-3	3
5	ME-212	Product Manufacturing Technologies	2-0-3-3	3
6	ME-213	Engineering Thermodynamics	3-1-0-4	4
7	ME-100	Reverse Engineering	0-0-2-1	1
8	ME-205	Machine Drawing	1-0-4-3	3
9	ME-210P	Fluid Mechanics Lab	0-0-2-1	1
10	ME-215	Manufacturing Engineering 1	3-0-0-3	3
11	ME-303	Heat Transfer	2.5-0.5-0-3	3
12	ME-307	Energy Conversion Devices	3-0-0-3	3
13	ME-303P	Heat Transfer Lab	0-0-3-1	1
14	ME-305	Design of Machine Elements	3-1-0-4	4
15	ME-309	Theory of Machines	3-1-0-4	4
16	ME-310	System Dynamics and Control	3-0-0-3	3
17	ME-311P	Design Lab 1	0-0-2-1	1
18	ME-315	Manufacturing Engineering 2	3-0-0-3	3
19	ME-312P	Design Lab 2	0-0-2-1	1
Total				50

Table 3: B. Tech Mechanical Engineering — B23 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket (choose 1)	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
6		IC-136	Understanding Biotechnology and its Applications	3-0-0-3	
7		IC-230	Environmental Science	3-0-0-3	
8	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
9		HS-342	German I	3-0-0-3	
10		HS-108	Basic English for Engineers	3-0-0-3	
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 4: B. Tech Mechanical Engineering — B23 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
9	IKS + HSS (choose 0–3 credits)	IC-181	Indian Knowledge Systems	2-0-2-3	0–3
10		HS-151	Introduction to English Literature	3-0-0-3	
11		HS-357	French Language and Culture	2-0-0-2	
12		HS-308	European Literature	2-0-0-2	
Total					22–25

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 5: B. Tech Mechanical Engineering — B23 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-272	Machine Learning	2-0-2-3	3
2	DC	EE-261	Electrical Systems Around Us	3-0-0-3	3
3	DC	IC-241	Material Science for Engineers	3-0-0-3	3
4	DC	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
5	DC	ME-210	Fluid Mechanics	2.5-0.5-0-3	3
6	DC	ME-212	Product Manufacturing Technologies	2-0-3-3	3
7	DC	ME-213	Engineering Thermodynamics	3-1-0-4	4
Total					22

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 6: B. Tech Mechanical Engineering — B23 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	DC	ME-100	Reverse Engineering	0-0-2-1	1
3	DC	ME-205	Machine Drawing	1-0-4-3	3
4	DC	ME-210P	Fluid Mechanics Lab	0-0-2-1	1
5	DC	ME-215	Manufacturing Engineering 1	3-0-0-3	3
6	DC	ME-303	Heat Transfer	2.5-0.5-0-3	3
7	DC	ME-307	Energy Conversion Devices	3-0-0-3	3
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 7: B. Tech Mechanical Engineering — B23 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	ME-303P	Heat Transfer Lab	0-0-3-1	1
2	DC	ME-305	Design of Machine Elements	3-1-0-4	4
3	DC	ME-309	Theory of Machines	3-1-0-4	4
4	DC	ME-310	System Dynamics and Control	3-0-0-3	3
5	DC	ME-311P	Design Lab 1	0-0-2-1	1
6	DC	ME-315	Manufacturing Engineering 2	3-0-0-3	3
Total					16

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 8: B. Tech Mechanical Engineering — B23 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	ME-312P	Design Lab 2	0-0-2-1	1
2	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					5

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 9: B. Tech Mechanical Engineering — B23 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	ME-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 10: B. Tech Mechanical Engineering — B23 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	ME-010	Internship		2
2	MTP	ME-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 16 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 11: Mechanical Engineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	ME-451	Refrigeration and Air Conditioning	3
2	ME-452	Robot Kinematics, Dynamics and Control	4
3	ME-501	Nanomanufacturing	3
4	ME-502	Functional Materials	3
5	ME-503	Micro/Nano Scale heat transfer	3
6	ME-504	Numerical Methods for Engineering Computation	3
7	ME-505	Applied Finite Element Method	3
8	ME-506	Fundamentals of Fracture Mechanics	3
9	ME-507	Micro and Nanoscale Fluid Mechanics	3
10	ME-508	Fundamentals of project management	3

Mechanical Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
11	ME-509	Nano-manufacturing	3
12	ME-510	Advanced Manufacturing Processes	3
13	ME-511	Manufacturing of composites	3
14	ME-513	Finite Element Methods in Engineering	4
15	ME-514	Fundamentals of Multiphase Flow	3
16	ME-516	Polymer Technology for Engineers	3
17	ME-517	Advanced Analytical Techniques for Engineers	4
18	ME-518	Conduction and Radiation	3
19	ME-519	Technical communication for Engineers	2
20	ME-520	Microwave based Manufacturing Processes	3
21	ME-521P	Laboratory course on Vehicle Design and Control	1
22	ME-522	High-Performance Scientific Computing	3
23	ME-523	Product Design	3
24	ME-524	Additive Manufacturing	3
25	ME-526	Programming Paradigm for Open-Source Software	1
26	ME-528	Thermal Management of Electronics	3
27	ME-530	Continuum Mechanics	3
28	ME-595	Experiential Learning in Composite Design, Manufacturing, Testing and Recycling	3
29	ME-600	Research Methodology	1
30	ME-601	Advanced Finite Element Methods	3
31	ME-602	Mechanical Vibration	3
32	ME-603	Advanced Fluid Mechanics	3
33	ME-604	Experimental Methods in Thermal Engineering	3
34	ME-605	Air Conditioning and Ventilation	3
35	ME-606	Advanced Solid Mechanics	3
36	ME-607	Materials Science For Failure Analysis	3
37	ME-608	Thin Films And Devices	3
38	ME-609	Functional Materials	3
39	ME-610	Advanced Thermodynamics	4
40	ME-611	Design and Optimization of Thermal Systems	3
41	ME-612	Introduction to Bio-materials	3
42	ME-613	Thermal Radiation	3
43	ME-614	Compressible Flow and Gas Dynamics	3
44	ME-615	Applied Computational Fluid Dynamics	3
45	ME-616	Dielectrics and Related Materials	3
46	ME-617	Mechanics of Composite Materials	3
47	ME-618	Stealth Technology: Infrared Signatures	3
48	ME-619	Experiments in Materials Science	3
49	ME-620	Modeling and Simulation	3
50	ME-621	Aircraft Propulsion	3

Mechanical Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
51	ME-622	Biomechanics of Musculoskeletal System	3
52	ME-625	Introduction to Turbulence and its Modeling	3
53	ME-626	Acoustics	3
54	ME-627	Mesh Independent Computational Techniques	3
55	ME-628	Impact Mechanics	3
56	ME-630	Machine learning for engineers	3
57	ME-631	Heat Transfer and Fluid Flow in Energy Systems	3
58	ME-632	Mechanics for Energy Systems	3
59	ME-633	Design of Energy Systems	3
60	ME-634	Thermodynamics for Energy Systems	3
61	ME-635	Manufacturing for Energy Systems	3
62	ME-636	Combustion Technology	3
63	ME-637	Wind Power Plant	3
64	ME-638	Solar Thermal Power Plant	3
65	ME-639	Thermal Power Plant Engineering	3
66	ME-640	Solar Power Utilization	3
67	ME-641	Finite Element Method	3
68	MT-201	Physics of Solids	3
69	MT-202	Applied Quantum Mechanics	3
70	MT-203	Materials Synthesis & Characterization	4
71	MT-204	Thermodynamics and Kinetics of Materials	3
72	MT-205	Functional Properties of Materials	4
73	MT-206	Extraction and Materials Processing	4
74	MT-301	Phase Transformations	3
75	MT-302	Transport Phenomena	3
76	MT-303	Computational Materials Science	4
77	MT-304	Mechanical Behavior of Materials	4
78	MT-501	Energy Conversion and Storage Technologies	3
79	MT-502	Recycling and Circular Economy	3
80	MT-503	Semiconductor Materials and Devices	3
81	MT-504	Powder metallurgical processing of Materials	3
82	MT-505	Thin Film Technology	3
83	MT-506	Biomaterials	3
84	MT-507	Modelling and Simulations in Material Science	3
85	MT-508	Iron and Steel Making	3
86	MT-509	Hydrogen Energy	3
87	MT-510	Colloids and Interfaces	3
88	MT-511	Sensor Materials and Technologies	3
<i>Compulsory project components — counted as DE, not selectable</i>			
89	ME-498P	Major Technical Project - I (compulsory; counted as DE)	4

Mechanical Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
90	ME-499P	Major Technical Project - II (compulsory; counted as DE)	4